

# Zhaoyi Xu

[Homepage](#) | [xu-zhaoyi](#) | [Google Scholar](#) | [in zhaoyi](#) | [zhaoyix@umich.edu](mailto:zhaoyix@umich.edu)

## ABOUT ME

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I am currently a PhD student in Mechanical Engineering at the University of Michigan, Ann Arbor, under the supervision of Professor Jianping Fu. My research lies at the intersection of biology and engineering, with a focus on understanding human development and health. Specifically, I employ human pluripotent stem cell (hPSC)-based organoids to model early human developmental processes. In addition, I am interested in inverse problems and medical image analysis, with an emphasis on developing interpretable AI models for healthcare applications.

## RESEARCH INTERESTS

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- Human developmental biology
- Human pluripotent stem cell (hPSC) models and therapies
- Interpretable/Reliable AI models for health

## EDUCATION

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**PhD, Mechanical Engineering, University of Michigan** 2021 - 2026 (expected)

- GPA: 4.0 / 4.0
- Advisor: Prof. [Jianping Fu](#)
- Thesis committee: [Cheri X. Deng](#), [Qiong Yang](#), and [Chengzhi Shi](#)

**MASc, Mechanical Engineering, University of Toronto** 2021

- GPA: 3.9 / 4.0
- Advisor: Prof. [Xinyu Liu](#)
- Thesis: An Improved Approach for Optogenetic Locomotion Control of *Caenorhabditis Elegans*

**BEng, Mechanical Engineering, Hong Kong University of Science and Technology** 2019

- The Dean's List, First Class Honours

**Exchange student in Mechanical Engineering, National University of Singapore** 2018

## SELECTED RESEARCH EXPERIENCE

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**Synthetic 3D hPSC Models on Peri-Implantation Development** 2021 - present  
University of Michigan, Supervisor: Professor Jianping Fu

- Developed biomimetic implantation-like niches for 3D hPSC organoid culture using microfluidics and microfabrication, replicating maternal tissue properties to study early human embryo development events.
- Independently initiated and executed three sub-projects, leading to publications [\[1\]](#) [\[8\]](#) [\[10\]](#).

## Neural Differentiation of hPSCs via Acoustic Tweezing Cytometry 2022 - 2024

University of Michigan, Supervised by Professor Jianping Fu, Professor Cheri X. Deng

- Utilized the Acoustic Tweezing Cytometry (ATC) technology to study hPSCs' mechanosensitive properties, for large-scale production of functional motor neurons for medical applications.
- Supported by NIH R01 GM143297 and participated as a core member, as main contributor to two annual progress reports, resulting in publication [4] [9].

## An Improved Approach for Optogenetic Locomotion Control of Caenorhabditis Elegans 2019 - 2021

University of Toronto, Supervised by Professor Xinyu Liu

- Collaborated on developing the living soft microrobot, RoboWorm, using optogenetic control of *C. elegans* with microfluidic tools; resulted in a notable Science Robotics publication [5].
- My work improved RoboWorm technology, elevating the success rate from 7.1% to 80.77%, by creating a new transgenic *C. elegans* strain and enhancing the control system's robustness.
- Collaborated closely with the renowned molecular biology team at Mount Sinai Hospital (Zhen lab) on a daily basis, gaining valuable bio-laboratory expertise.

## Gecko-Inspired Symbiosis Robotics End-Effector 2018 - 2019

HKUST Robotics Institute, Supervised by Professor Michael Yu Wang

- This is my undergraduate Final-Year Project, where I utilized 3D-printing and Finite Element Method, designed and microfabricated a gecko-inspired gripper for delicate object handling in industrial production.
- Devoted 300+ hours, led a group of 4 students and awarded the HKUST President's Cup for the most outstanding final-year projects of the year.

## 3D-Printed Devices with Negative Poisson's Ratio 2018 - 2019

NUS Centre for Additive Manufacturing (AM.NUS), Supervised by Professor Wen Feng Lu

- Worked with Dr. Lu, director of AM.NUS and his team, we developed medical protective equipment and soft robots with auxetic structures using 3D-Bioprinting (publication [6] [11]).

## PUBLICATIONS

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### Journals

- [1] **Zhaoyi Xu**, Weiping Li, Xufeng Xue, Nicholas G. Schott, Ruxin Yang, Shiyu Sun, Jeyoon Bok, Jan P. Stegmann, Jianping Fu, and Cheri X. Deng, "Acoustic cell patterning reveals geometry- and substrate-dependent vasculogenesis and human embryo model development," *Nature Communications*, under revision, 2026.
- [2] Tao Hong, **Zhaoyi Xu**, Se Young Chun, Luis Hernandez-Garcia, and Jeffrey A. Fessler\*, "Convergent complex quasi-Newton proximal methods for gradient-driven denoisers in compressed sensing MRI reconstruction," *IEEE Trans. on Computational Imaging*, vol. 11, pp. 1534–1547, 2025. DOI: [10.1109/TCI.2025.3625052](https://doi.org/10.1109/TCI.2025.3625052).
- [3] Tao Hong, **Zhaoyi Xu**, Jason Hu, and Jeffrey A. Fessler\*, "Using randomized Nyström preconditioners to accelerate variational image reconstruction," *IEEE Trans. on Computational Imaging*, vol. 11, pp. 1630–1643, 2025. DOI: [10.1109/TCI.2025.3622903](https://doi.org/10.1109/TCI.2025.3622903).
- [4] **Zhaoyi Xu**, Shiyong Liu, Xufeng Xue, Weiping Li, Jianping Fu\*, and Cheri X Deng\*, "Microbubble behaviors in acoustic tweezing cytometry regulates embryonic stem cell differentiation," *Scientific Reports*, vol. 13, no. 18030, 2023. DOI: [10.1038/s41598-023-45397-5](https://doi.org/10.1038/s41598-023-45397-5).

- [5] Xianke Dong, Sina Kheiri, Yangning Lu, **Zhaoyi Xu**, Mei Zhen, and Xinyu Liu\*, “Towards a living soft microrobot: Optogenetic locomotion control of caenorhabditis elegans,” *Science Robotics*, vol. 6, no. 55, eabe3950, 2021. DOI: [10.1126/scirobotics.abe3950](https://doi.org/10.1126/scirobotics.abe3950).
- [6] Mingcan Liu, **Zhaoyi Xu**, Jing Jie Ong, Jian Zhu, and Wen Feng Lu\*, “An earthworm-like soft robot with integration of single pneumatic actuator and cellular structures for peristaltic motion,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 7840–7845, 2020. DOI: [10.1109/IROS45743.2020.9341166](https://doi.org/10.1109/IROS45743.2020.9341166).
- [7] Sihan Huang, **Zhaoyi Xu**, Guoxin Wang\*, Cong Zeng, and Yan Yan, “Reconfigurable machine tools design for multi-part families,” *The International Journal of Advanced Manufacturing Technology*, vol. 105, no. 1, pp. 813–829, 2019. DOI: [10.1007/s00170-019-04236-6](https://doi.org/10.1007/s00170-019-04236-6).

## Conferences

- [8] **Zhaoyi Xu**, Weiping Li, Jianping Fu\*, and Cheri X Deng\*, “Controlled development of a human epiblast model using acoustic patterning,” 8th Bioengineering and Translational Medicine Conference by AIChE, San Diego, US, 2024.
- [9] **Zhaoyi Xu**, Weiping Li, Jianping Fu\*, and Cheri X Deng\*, “Formation and development of a human epiblast model in biomimetic implantation-like niche through acoustic hpscs patterning,” The Great Lakes Developmental Biology Meeting, Toronto, Canada, 2023.
- [10] **Zhaoyi Xu**, Xufeng Xue, and Jianping Fu\*, “Probing mechanobiology of human neural crest cell development using a patterned microfluidic neural tube model,” Keystone Symposia on Molecular and Cellular Biology, Keystone, CO, United States, 2022.
- [11] Yafeng Han, **Zhaoyi Xu**, and Wen Feng Lu\*, “Design of conforming surface for human skin at highly-stretched joint areas with additive manufacturing,” 5th International Conference on Additive Manufacturing and Bio-Manufacturing (ICAM-BM), Beijing, China, 2018.

## HONORS AND AWARDS

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|---------|----------------------------------------------------------|--------------------------|
| 2024    | 8th Bioengineering and Translational Medicine Conference | Young Investigator Award |
| 2021-23 | UMich Rackham Travel Grant                               | USD 2,150                |
| 2021    | UMich ME Department Fellowship                           | USD 89,775               |
| 2019-21 | UofT MIE Department Fellowship                           | Full Scholarship         |
| 2021    | UofT Barbara and Frank Milligan Fellowships              | CAD 10,000               |
| 2020    | UofT MIE Endowed Fellowship                              | CAD 3,000                |
| 2019    | HKUST President’s Cup                                    | Gold Award               |
| 2018    | HKSAR Government Talent Development Scholarship          | HKD 10,000               |
| 2016    | HKUST Robot Design Contest                               | Champion                 |
| 2013    | China National Awarding Program of Future Scientists     | Top 100                  |
| 2009    | China Arts Grade Examination - Piano, Grade 10           | Top Level                |

## TEACHING EXPERIENCE

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### ME 335 Heat Transfer

University of Michigan, Winter 2026

Graduate Student Instructor (Teaching Assistant) (20 hrs/week)

- Led instruction for 93 students; designed/delivered recitation lectures and held office hours.
- Developed experience in large-class teaching and course coordination.

### ME 495 Laboratory II

University of Michigan, Fall 2025

Graduate Student Instructor (Teaching Assistant) (20 hrs/week)

- Designed and taught two lab sections (24 students) for senior-level laboratory course.
- Broad topics included heat transfer, mechanical dynamics, and AFM-based nanoscale metrology.
- Gained extensive self-directed learning and close student mentorship.

### MIE 342 Circuits with Applications to Mechanical Engineering Systems

University of Toronto, Fall 2020

Teaching Assistant (15 hrs/week)

- Supported a lab course of  $\approx 130$  students in a four-TA team.
- Co-developed online laboratory tools for remote instruction during COVID-19, enabling virtual circuit assembly and experimental data acquisition.

## STUDENT MENTORING

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- **Ruxin Yang** (Aug. 2024 – Sep. 2025), M.S. student, Biomedical Engineering, University of Michigan. Topic: Acoustic cell patterning.
- **Linyuan Li** (Jun. 2025 – Sep. 2025), B.S. student, Biochemistry, University of Michigan. Topic: Mechanical regulation of zebrafish PSM cells using geometrically modulated elastomeric substrates.

## SKILLS

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### Laboratory

**Cell biology:** Human Pluripotent Stem Cell • Imaging and Microscopy • Immunostaining • qPCR • Single-cell RNA Sequencing

**Engineering:** Microfluidics • Nanofabrication • 3D Printing/Bioprinting

**Quantitative Analysis:** Python • R • Matlab • C/C++

**Illustration:** ImageJ • Adobe Photoshop • Adobe Illustrator • Procreate

**Office:** Office Suite • Google Workspace •  $\text{\LaTeX}$

**Teamwork:** Multi-tasking • Interpersonal skills

**Languages:** English • Mandarin • Cantonese